

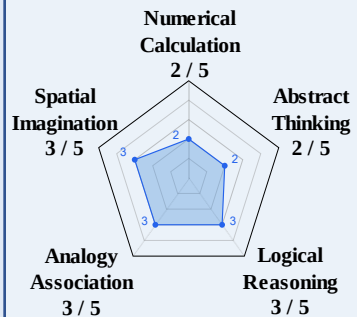
Input

Global Teaching Goal



Algebra Goal:
linear equations
with one variable.

Student Profile (Tree + Scores)

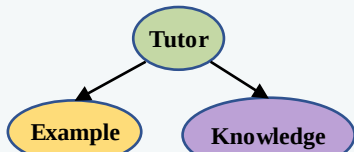


Dataset

Algebra Dataset:
Pen costs \$3. Total \$5
for 1 pen and 2
erasers. Cost of 1
eraser?
 $2x + 3 = 5$

Tree-Based Local Search Stage

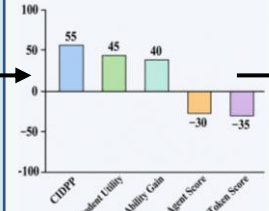
Structure {i}



Lesson Plan

Knowledge:
linear equation
Example:
 $4x+2=10$

Performance

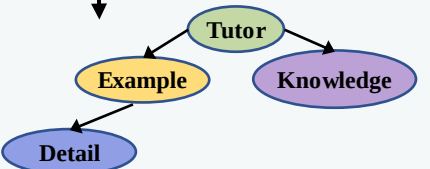


Error Analysis

The steps are
not specific
enough

Proposal: Create a new agent for details.

Structure {i+1}



Lesson Plan

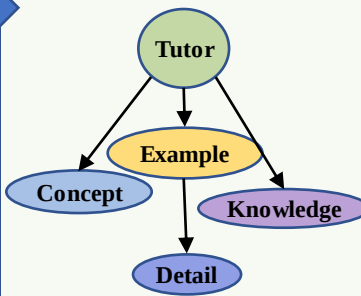
Knowledge:
linear equation
Example:
 $4x+2=10$
Detail:
Explain step by step

Performance

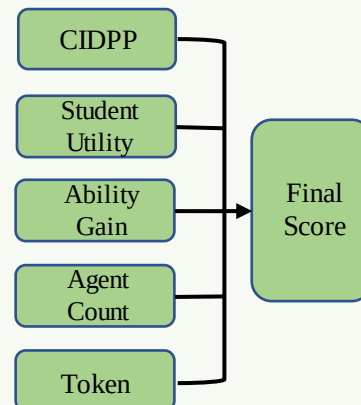


Transition

Best Tree Structure



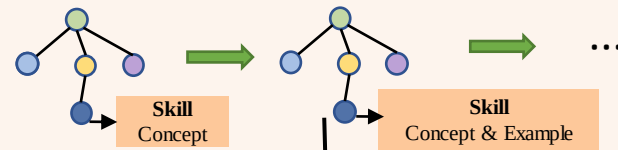
Evaluator



Graph-Based Global Search Stage

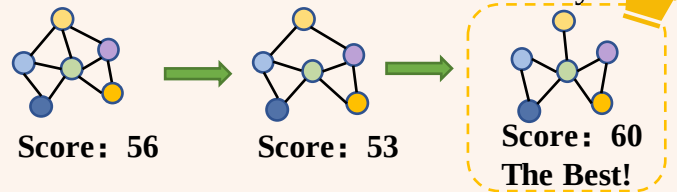
1.Skill Evolution

Improve agent skills

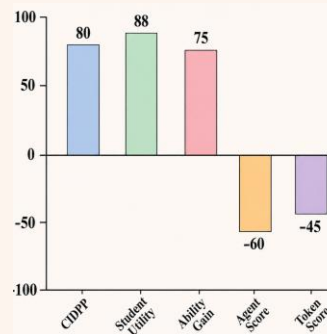


2. Structure Iteration

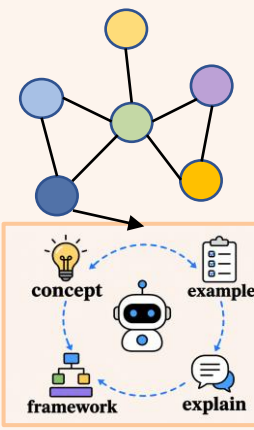
Mutate structure for diversity



Performance

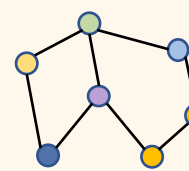


New Structure



Output layer

Optimized structure



Final Lesson Plan

Knowledge: linear equations
basics
Concept: $ax + b = c$ form
Example: $3x + 5 = 20$
Detail: isolate variable steps
Guide: practice and extension
problems

New Student Profile

